Do Now · retrieval

Graphs and graph representations

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Where is a value 8 inserted if BST root 10 has no left child?

Q2

What is the purpose of a next pointer in a linked list?

Q3

What does zero-based indexing mean?

Q4

In a BST root 20, left 10, right 30, where does 25 go?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify vertices — A, B, C and D are graph vertices.

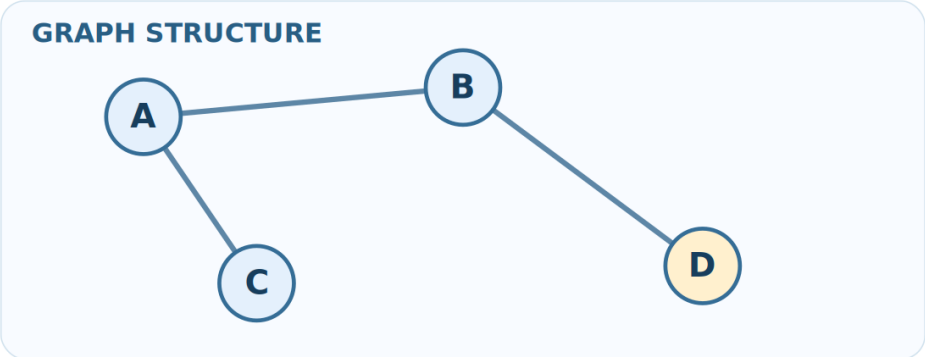
Core idea 2

Follow edges — Edges connect A–B, A–C and B–D. Because this graph is undirected, each connection works both ways.

Core idea 3

Write adjacency lists — A:B,C; B:A,D; C:A; D:B. Every undirected edge appears in both endpoint lists.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

vertex — A node in a graph.

edge — Connection between vertices.

adjacency list — Representation listing neighbours of each vertex.

adjacency matrix — Two-dimensional matrix marking which vertices are connected.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

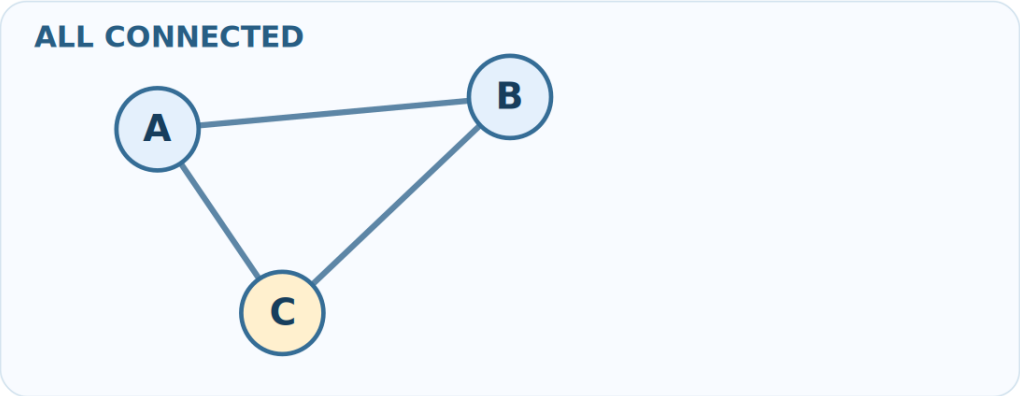
Build adjacency lists for triangle A–B, B–C, A–C.

Follow the worked method

1. Predict — Every vertex connects to the other two vertices.

2. Confirm and explain — A:B,C; B:A,C; C:A,B. Each undirected edge is listed from both endpoints.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

A is adjacent to B; D is adjacent to C.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Name the vertices and number of edges in A–B, A–C, C–D.

Question 2

Give adjacency lists for an undirected graph with edges AB, BC, CD.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare adjacency matrix and adjacency list storage on a sparse graph with many vertices.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

In an undirected graph, A–B is represented in both A's and B's adjacency lists.

Use key vocabulary from memory

1. Explain vertex in your own words.

2. Explain edge in your own words.

3. Explain adjacency list in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.